Indian Institute of Technology Kharagpur

3D Dental Scan Segmentation and Labeling Using Deep Learning

Internship Report

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Abstract

This report presents the implementation and evaluation of deep learning models for automated 3D dental scan segmentation and labeling. The work is based on the solutions of the MICCAI 2022 3D Teeth Scan Segmentation and Labeling Challenge, and evaluates three models: TGNet (Tooth Group Network), TSegNet (Tooth Segmentation Network), and Dilated TSegNet. TGNet and TSegNet were fully implemented and trained on a dataset of 1200 dental scans from 600 patients, following a two-stage training pipeline — coarse tooth grouping followed by boundary or centroid refinement — with results evaluated on a held-out test set of 120 scans using standard MICCAI 2022 challenge metrics (IoU, F1, and semantic accuracy). Qualitative per-vertex failure-case analysis is presented for all three models to characterize common error patterns, particularly around posterior tooth segmentation. Dilated TSegNet was separately trained and evaluated on the official Teeth3DS dataset split, and its results are reported independently since they use a different evaluation protocol. The complete training pipeline was set up on a remote GPU server, including environment configuration, data preprocessing, model training, and evaluation.

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1 Introduction

Dental diagnosis and treatment planning increasingly rely on 3D intraoral scans that capture the geometry of a patient's teeth and gums with high precision. Automated segmentation and labeling of individual teeth from these scans is a clinically important task that can significantly reduce the manual effort required by dental professionals.

This internship project focuses on implementing, training, and evaluating deep learning models for 3D tooth segmentation. The work is based on the repository by Lim and Kim [1], which won first place at the MICCAI 2022 3D Teeth Scan Segmentation and Labeling Challenge [2].

Figure 1 shows a representative example of the target output for this task: each tooth on the dental arch is assigned its own instance color and FDI label, while the surrounding gingiva is left unlabeled. Reproducing this kind of per-tooth, per-vertex prediction is the central goal of the models implemented and evaluated in this report.

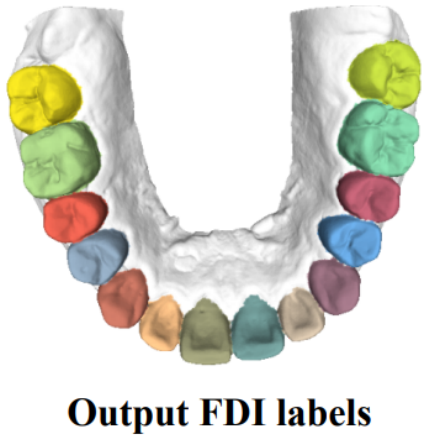


Figure 1: Example output showing predicted FDI tooth labels on a lower jaw scan, with each tooth instance colored individually. Reproduced from Ben-Hamadou et al. [2].

1.1 Problem Statement

Given a 3D dental mesh scan in `.obj` format, the task is to:

- Assign each vertex a **semantic label** (which tooth it belongs to, numbered 1–16, or gingiva as class 0)
- Assign each vertex an **instance label** (which specific tooth instance it belongs to)

1.2 Objectives

- Set up a complete deep learning training pipeline on a remote GPU server

- Implement and train TGNet and TSegNet models
- Evaluate and compare model performance on a held-out test set
- Document results for comparison with additional baseline models

2 Environment Setup

2.1 Hardware and Software

All experiments were conducted on a remote GPU server accessed via SSH using MobaXterm. File transfers were performed using WinSCP. Table 1 summarizes the hardware and software environment.

Table 1: Hardware and Software Environment

Component	**Details**
GPU	NVIDIA GeForce (10.76 GiB VRAM)
Operating System	Ubuntu 18.04.5 LTS (x86_64)
CUDA Version	11.3
Python Version	3.7
PyTorch Version	1.10.1
Environment Manager	Miniconda 4.12.0
Remote Terminal	MobaXterm (SSH)
File Transfer	WinSCP (SFTP)
Process Manager	GNU Screen

2.2 Installation Challenges

Several technical challenges were encountered and resolved during environment setup, as summarized in Table 2.

Table 2: Installation Challenges and Solutions

Challenge	**Solution**
GLIBC 2.27 incompatible with latest Miniconda	Used older Miniconda 4.12.0 (compatible with GLIBC 2.17+)
`THC/THC.h` header missing with PyTorch 1.12.1	Downgraded to PyTorch 1.10.1 which retains the header
Custom `pointops` CUDA library build failure	Resolved by matching PyTorch version

3 Dataset

3.1 Dataset Description

The dataset used is from the MICCAI 2022 3D Teeth Scan Segmentation and Labeling Challenge [2]. It consists of 3D intraoral dental scans in `.obj` format with corresponding ground truth labels in `.json` format.

Table 3: Dataset Summary

Property	**Details**
Total Patients	600
Total Scans	1200 (upper + lower jaw per patient)
File Format	`.obj` (mesh) + `.json` (labels)
Number of Classes	17 (16 teeth + gingiva)

3.2 Data Split

The dataset was split into training, validation, and test sets at the patient level to prevent data leakage, as shown in Table 4.

Table 4: Train/Validation/Test Split

Split	**Ratio**	**Patients**	**Scans**
Train	80%	480	960
Validation	10%	60	120
Test	10%	60	120
Total	100%	**600**	**1200**

3.3 Data Preprocessing

Raw mesh files were preprocessed before training using the following pipeline:

1. Load `.obj` mesh and corresponding `.json` label file
2. Compute vertex normals for each mesh
3. Normalize 3D coordinates using global Y-axis min/max values
4. Apply **Farthest Point Sampling (FPS)** to downsample meshes to 24,000 points (if mesh has more than 24,000 vertices)
5. Save preprocessed data as `.npy` files

A total of 1200 `.npy` files were generated from the preprocessing pipeline.

3.4 Data Augmentation

The following augmentations were applied during training to improve generalization:

- **Random Scaling:** Scale factor sampled uniformly from [0.85, 1.15]
- **Random Rotation:** Angle sampled from $[-30, 30]$ around fixed axis
- **Random Translation:** Offset sampled from $[-0.2, 0.2]$

4 Methodology

4.1 TGNet — Tooth Group Network

TGNet is a two-stage point cloud segmentation network proposed by Lim and Kim [1]. It was the winning method at the MICCAI 2022 challenge.

4.1.1 Stage 1: FPS Grouping Network (`tgnet_fps`)

The first stage uses a Farthest Point Sampling (FPS) based grouping network to segment the point cloud into individual tooth regions. It learns to predict:

- Tooth class labels (semantic segmentation)
- Point offsets toward tooth centroids
- Boundary regions between teeth

4.1.2 Stage 2: Boundary-Aware Grouping Network (`tgnet_bdl`)

The second stage refines the segmentation using a boundary-aware point sampling strategy. It loads the trained FPS model and further refines tooth boundaries using a boundary loss term.

Loss Functions:

- `tooth_class_loss`: Cross-entropy loss for tooth classification
- `offset_loss`: L2 loss for centroid offset prediction
- `chamf_loss`: Chamfer distance loss
- `cbl_loss`: Boundary loss for sharp boundary prediction

4.2 TSegNet — Tooth Segmentation Network

TSegNet [3] is a two-stage architecture for instance segmentation of teeth.

4.2.1 Stage 1: Centroid Prediction Module

Trains a module to predict the centroid location of each tooth. Only centroid prediction losses are used in this stage:

- `dist_loss`: Distance prediction loss
- `cent_loss`: Centroid localization loss
- `chamf_loss`: Chamfer distance loss

4.2.2 Stage 2: Full Segmentation Module

Uses the pretrained centroid prediction module and adds segmentation losses:

- `seg_1_loss`: Binary segmentation loss (stage 1)
- `seg_2_loss`: Binary segmentation loss (stage 2)
- `id_pred_loss`: Tooth ID prediction loss

4.3 Training Configuration

Table 5: Training Hyperparameters

Parameter	**TGNet FPS**	**TGNet BDL**	**TSegNet**
Batch Size	1	1	1
Optimizer	SGD	SGD	SGD
Learning Rate	1e-1	1e-1	–
Momentum	0.9	0.9	0.9
Weight Decay	1e-4	1e-4	1e-4
LR Scheduler	Cosine	Cosine	Cosine
Min GPU RAM	11 GB	11 GB	11 GB
Training Time	∼3 days	∼1 day	∼3 days

5 Evaluation Metrics

Three standard metrics from the MICCAI 2022 challenge were used:

- **IoU (Intersection over Union / TSA):** Measures the overlap between predicted and ground truth tooth regions.

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{1}$$

- **F1 Score (TSA — Tooth Segmentation Accuracy):** Harmonic mean of precision and recall.

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

- **SEM_ACC (TIR — Tooth Identification Rate):** Measures the fraction of tooth instances correctly identified by their semantic class label.

6 Results

6.1 TGNet Results

TGNet was evaluated on 120 test scans (60 patients). Table 6 reports the mean and standard deviation of each metric across all test scans.

Table 6: TGNet Test Set Results (mean ± std over 120 scans)

Metric	**Mean ± Std**
IoU (TSA)	0.9194 ± 0.0378
F1 Score (TSA)	0.9537 ± 0.0285
SEM_ACC (TIR)	0.9541 ± 0.1150

6.2 TSegNet Results

TSegNet was evaluated on the same 120 test scans. Table 7 reports the mean and standard deviation of each metric across all test scans.

Table 7: TSegNet Test Set Results (mean ± std over 120 scans)

Metric	**Mean ± Std**
IoU (TSA)	0.9047 ± 0.0461
F1 Score (TSA)	0.9458 ± 0.0354
SEM_ACC (TIR)	0.9454 ± 0.1212

6.3 Comparison of TGNet and TSegNet

Table 8 compares the two models evaluated under the identical 120-scan test protocol described in Section 5.

Table 8: Model Comparison on Test Set (mean ± std, 120 scans)

Model	**IoU (TSA)**	**F1 (TSA)**	**SEM_ACC (TIR)**
TGNet (Ours)	$\mathbf{0.9194 \pm 0.0378}$	$\mathbf{0.9537 \pm 0.0285}$	$\mathbf{0.9541 \pm 0.1150}$
TSegNet	0.9047 ± 0.0461	0.9458 ± 0.0354	0.9454 ± 0.1212

6.4 Dilated TSegNet Results

Dilated TSegNet [4] was trained and evaluated separately, using the official Dilated TSegNet codebase's own train/test split rather than the 120-scan patient-level split used for TGNet and TSegNet above. Its reported metrics are also different: per-vertex accuracy and semantic mean IoU (a class-averaged measure), rather than the instance-level IoU (TSA), F1 (TSA), and SEM_ACC (TIR) metrics used for TGNet and TSegNet. For this reason, its results are kept separate from Table 8 rather than merged into it.

Table 9: Dilated TSegNet Results

Training Data	**Val. Accuracy**	**Val. mIoU**	**Val. Loss**
Partial dataset (1,200 scans / 600 patients)	0.9516	0.7834	0.2123
Full official Teeth3DS dataset (∼1,799 scans)	0.9489	0.8406	0.2135

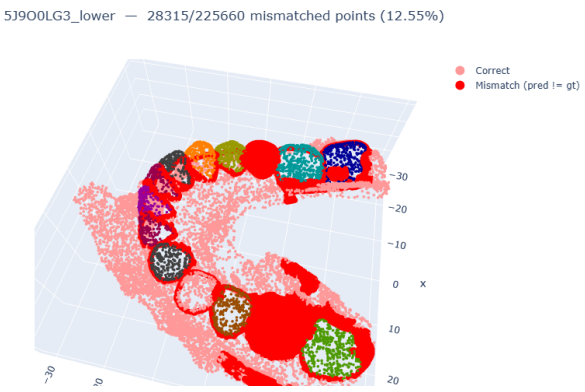
Moving from the partial dataset to the full official dataset improved mean IoU by 5.7 points (0.7834 → 0.8406) despite a marginal drop in per-vertex accuracy, suggesting that dataset completeness — rather than hyperparameter tuning — was the dominant factor limiting segmentation quality on harder or rarer cases.

6.5 Qualitative Results — Failure Case Analysis

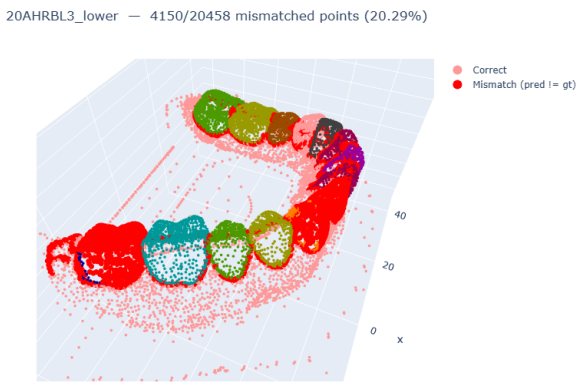
To complement the aggregate metrics in Tables 6–8, per-vertex prediction vs. ground-truth comparisons were generated for individual test scans. Each figure below highlights mismatched vertices (where the predicted label disagrees with ground truth) against correctly labeled vertices, with the mismatch rate reported in each subcaption. Cases are ordered from best to worst (lowest to highest mismatch percentage) to illustrate the spectrum of model behavior, from near-perfect segmentation to clinically significant failures.

6.5.1 TSegNet Failure Cases

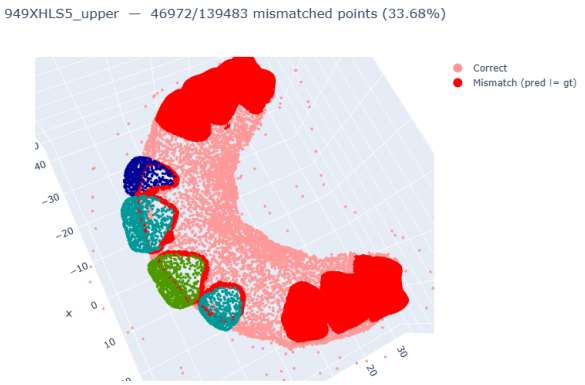
Figure 2 shows four representative TSegNet test scans spanning a wide range of mismatch rates (12.6%–43.8%).



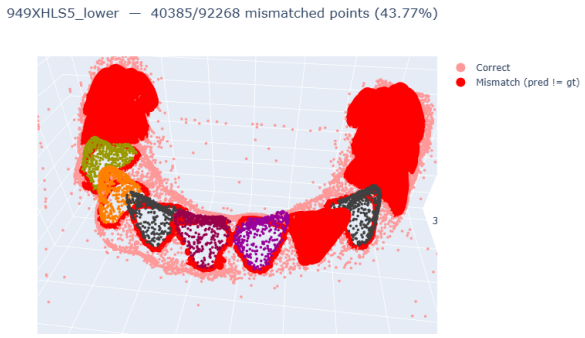
(a) 5J9O0LG3_lower — 12.55% mismatched



(b) 20AHRBL3_lower — 20.29% mismatched



(c) 949XHLS5_upper — 33.68% mismatched



(d) 949XHLS5_lower — 43.77% mismatched

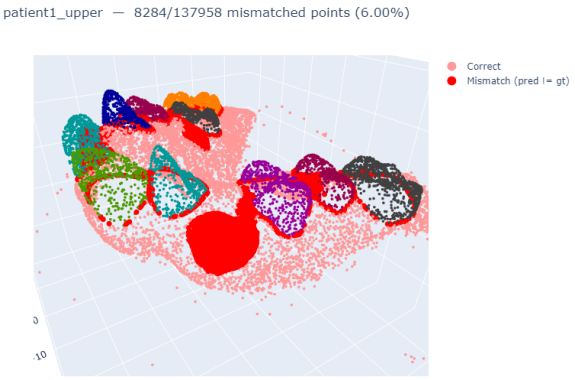
Figure 2: TSegNet qualitative results, ordered best to worst by mismatch rate.

In the best-performing case (Figure 2a), most individual teeth are cleanly separated into distinct instance colors, with errors confined to a single large mismatched region on the posterior right side, likely corresponding to a missed or merged molar instance. As the mismatch rate increases (Figure 2b), a similar pattern persists: the anterior teeth remain well segmented while a contiguous block of posterior points is mislabeled, suggesting the centroid prediction module is failing to localize molar instances correctly rather than producing scattered, random errors.

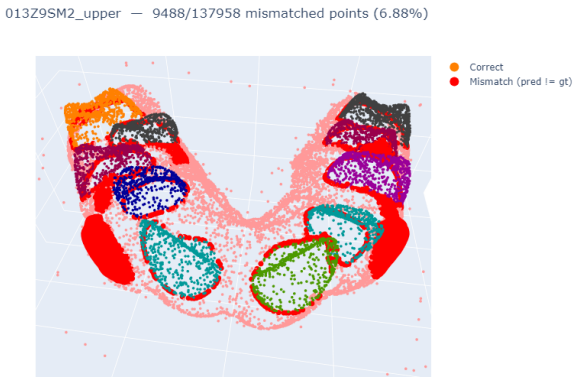
The two worst cases (Figures 2c and 2d), both from the same patient (949XHLS5), show a consistent and severe failure mode: **large contiguous red regions on both the left and right posterior segments of the arch**, with only the anterior teeth correctly identified. This indicates TSegNet's centroid prediction module struggles on this particular scan geometry, possibly due to crowding, partial occlusion of molars, or an atypical jaw curvature that places posterior centroids outside the distribution seen during training. The fact that *both* the upper and lower jaw of the same patient exhibit this failure pattern suggests a per-patient/per-scan difficulty (e.g., scan quality or anatomy) rather than a purely random model error.

6.5.2 TGNet Failure Cases

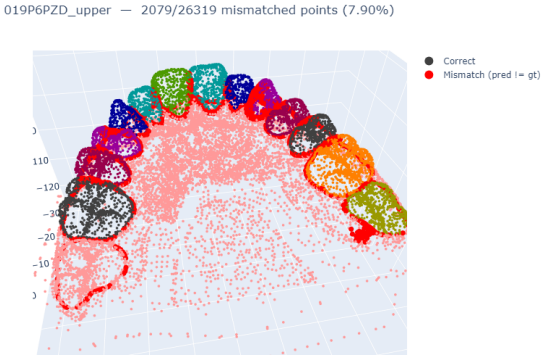
Figure 3 shows five representative TGNet test scans spanning mismatch rates from 6.00% to 37.78%.



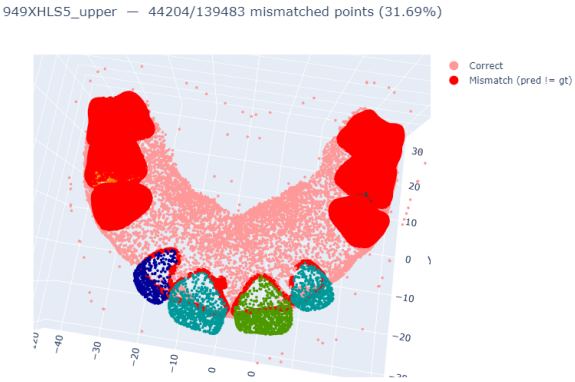
(a) patient1_upper — 6.00% mismatched



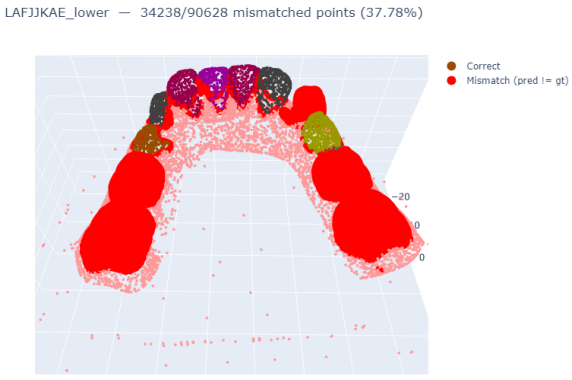
(b) 013Z9SM2_upper — 6.88% mismatched



(c) 019P6PZD_upper — 7.90% mismatched



(d) 949XHLS5_upper — 31.69% mismatched



(e) LAFJJKAE_lower — 37.78% mismatched

Figure 3: TGNet qualitative results, ordered best to worst by mismatch rate.

The three best TGNet cases (Figures 3a–3c) show errors concentrated almost entirely at the **boundary ring around each tooth** and along the gingiva-to-arch transition at the two ends of the jaw, rather than entire teeth being missed. This is consistent with

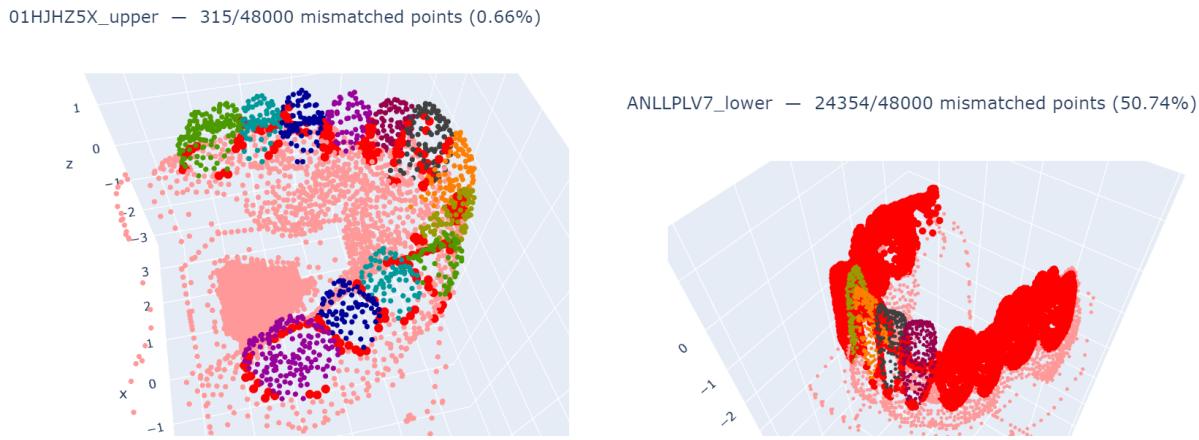
the boundary-aware refinement stage (`tgnet_bdl`) successfully separating tooth instances, with residual error limited to a thin margin of ambiguous boundary points — a far less severe failure mode than the large contiguous blocks seen in the worst cases.

The two worst TGNet cases (Figures 3d and 3e) again show large, solid mismatched regions covering entire posterior segments of the arch, with only four to six anterior teeth correctly segmented. Notably, **949XHLS5_upper appears as a failure case for both TSegNet and TGNet** (Figures 2c and 3d), with strikingly similar error regions in both models. This cross-model agreement strengthens the hypothesis that this particular scan presents an inherently difficult case — likely due to missing posterior teeth, unusual molar crowding, or mesh artifacts in the raw scan — rather than a weakness specific to either architecture.

Common failure pattern. Across both models, posterior (molar) regions are disproportionately responsible for high mismatch rates, while anterior teeth are consistently well segmented even in otherwise poor scans. This suggests future work should prioritize improving centroid localization and boundary refinement specifically for posterior teeth, which may benefit from additional training data emphasizing molar crowding and partial occlusion cases.

6.5.3 Dilated TSegNet Failure Cases

Figure 4 shows one best-case and one worst-case Dilated TSegNet test scan, selected from its own official Teeth3DS test split (see Section 6.4 for why this is a different test set than the one used for TGNet and TSegNet above).



(a) 01HJHZ5X_upper — 0.66% mismatched (b) ANLLPLV7_lower — 50.74% mismatched

Figure 4: Dilated TSegNet qualitative results: best case (left) and worst case (right) by mismatch rate, on Dilated TSegNet's own test split.

In the best case (Figure 4a), nearly every tooth is cleanly separated with the correct class color, and the small number of mismatched (red) points are concentrated

along tooth-to-tooth boundaries rather than whole-tooth misclassifications — the same boundary-ambiguity pattern observed in TGNet's and TSegNet's best cases. In the worst case (Figure 4b), the failure mode is markedly different in character from TGNet/TSegNet's worst cases: rather than isolated posterior regions being mislabeled, an entire contiguous arc of the arch (roughly half the visible teeth) is replaced by solid red, with only one side of the arch retaining distinct, correctly colored tooth clusters. This is consistent with a per-vertex semantic segmentation failure across a whole spatial region of the scan, rather than the more localized instance-grouping failures seen in TGNet and TSegNet's centroid- and boundary-based approaches — a difference plausibly attributable to the architectural gap between Dilated TSegNet's semantic-only formulation and the other two models' instance-aware designs.

7 Discussion

TGNet achieved a mean IoU of 0.9194 ± 0.0378 and mean F1 of 0.9537 ± 0.0285 across 120 test scans, comparable to the authors' published results in the MICCAI 2022 challenge. TSegNet achieved a mean IoU of 0.9047 ± 0.0461 and mean F1 of 0.9458 ± 0.0354, performing slightly below TGNet on all three metrics. Both models exhibit relatively high standard deviations in SEM_ACC (± 0.1150 and ± 0.1212 respectively), indicating that semantic tooth identification varies considerably across patients rather than failing uniformly.

Several factors contributed to TGNet's stronger performance:

- The two-stage architecture (FPS + BDL) effectively separates coarse segmentation from boundary refinement
- FPS ensures uniform coverage of the point cloud
- The boundary-aware loss term sharpens inter-tooth boundaries
- Data augmentation improved generalization to unseen scans

A key practical challenge was the GPU memory constraint (10.76 GiB), which required batch size of 1 and careful process management to avoid out-of-memory errors.

Separately, Dilated TSegNet (Section 6.4) showed a 5.7-point mIoU improvement when trained on the full ∼1,799-scan official Teeth3DS dataset compared to a smaller 1,200-scan subset, underscoring that dataset completeness has a substantial effect on segmentation quality independent of architecture choice. This result uses a different test split and metric protocol than TGNet/TSegNet and is not directly comparable to the figures above, but is a useful data point for future dataset-scaling decisions in this line of work. The qualitative comparison in Section 6.5 further suggests Dilated TSegNet's

worst-case failures are more spatially contiguous and semantic in nature, in contrast to TGNet/TSegNet's more localized, instance-grouping failures — though this observation is based on a single worst-case example and would benefit from a larger qualitative sample before drawing firm conclusions.

8 Conclusion

This internship successfully implemented and evaluated deep learning models for 3D dental scan segmentation. The complete pipeline — from environment setup on a remote GPU server to data preprocessing, training, and evaluation — was established from scratch.

TGNet achieved a mean IoU of 0.9194 ± 0.0378 and mean F1 of 0.9537 ± 0.0285 over 120 test scans, demonstrating the effectiveness of the two-stage point cloud segmentation approach. TSegNet achieved a mean IoU of 0.9047 ± 0.0461 and mean F1 of 0.9458 ± 0.0354, performing competitively though slightly below TGNet across all metrics. The qualitative failure-case analysis in Section 6.5 revealed that both TGNet and TSegNet share a common weakness in posterior (molar) segmentation, with one test patient (949XHLS5) proving difficult for both architectures — suggesting scan-level difficulty rather than a model-specific limitation.

Dilated TSegNet was separately trained and evaluated on both a partial (1,200-scan) and the full ($\sim$1,799-scan) official Teeth3DS dataset, achieving validation mIoU of 0.7834 and 0.8406 respectively (Section 6.4), with qualitative best/worst-case examples presented in Section 6.5. These results use a different test split and metric protocol than TGNet/TSegNet, so they are reported separately rather than compared directly.

Future work could include:

- Re-evaluating Dilated TSegNet on the same 120-scan test split and metric protocol as TGNet/TSegNet, for a direct quantitative and qualitative comparison
- Training with wandb tracking for detailed loss curve analysis
- Testing on additional baseline models (PointNet, PointNet++, DGCNN)
- Extending TGNet/TSegNet training to the full official dataset, following the mIoU improvement observed with Dilated TSegNet
- Optimizing inference speed for clinical deployment

References

[1] Ho Yeon Lim and Min Chang Kim, *ToothGroupNetwork: Tooth Group Network for 3D Dental Scan Segmentation*, MICCAI 2022 3D Teeth Scan Segmentation and Labeling Challenge, 2022. https://github.com/limhoyeon/ToothGroupNetwork

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[4] L. Krenmayr et al., *DilatedToothSegNet: Tooth Segmentation Network using Dilated Edge Convolutions*, 2024. https://github.com/LucasKre/dilated_tooth_seg_net